

Table 1 Tools for designing DNA sequence libraries and for related sequence-design problems. Members of the final two rows are named in the table and cited individually.

Tool	Purpose / availability	Key features	Output
MPRA Design Tools [1]	Design MPRA libraries and estimate statistical power. R package (GitHub).	Barcode assignment with error-correcting sets; power analysis over effect size and sequencing depth	Barcoded constructs and statistical power estimates
MPRAnator [2]	Design MPRA libraries of motif arrangements and sequence variants. Web service.	Combinatorial placement of motifs; randomly sampled and combinatorial substitution sets; scrambled and reverse-complement controls	Barcoded library of sequences (FASTA)
Mutation Maker [3]	Design mutagenic oligos for building variant libraries in the laboratory. Web application (Docker); REST API.	Degenerate codons at user-listed sites; several substitutions combined in one reaction; de novo gene assembly with per-site variant frequencies	Primers, or gene fragments with the proportion of each variant to pool
Oligopool Calculator [4]	Add barcodes, primers, and spacers to a supplied variant set, and analyze the sequencing readout. Python package (PyPI); command line.	Constraint-aware barcodes, primers, and spacers; degenerate compression of a supplied variant set; off-target screening against a host genome	Synthesis-ready constructs; variant counts from sequencing reads
PoolParty	Design libraries combining multiple variant types, for laboratory assays or for probing genomic AI models. Python package (PyPI).	Libraries specified as a directed acyclic graph of operations, freely mixing mutagenesis schemes; inputs from sequences, FASTA, degenerate templates, or motifs; sequences generated only on request	Library of sequences, each paired with a record of how it was constructed
tangermeme [5]	Generate perturbed sequence sets to probe cis-regulatory logic in deep learning models. Python package (PyPI).	Saturation mutagenesis of input sequences; insertion and removal of motifs across sets of sequences	Perturbed sequences, model predictions, or attribution scores
VaLiAnT [6]	Design and annotate libraries for saturation genome editing and cDNA deep mutational scanning. Command-line tool (source, Docker).	Saturation mutagenesis from genomic coordinates; several mutation types per target region; transcript-aware, including codons split across exons	Per-region libraries with metadata tables and VCF carrying variant consequence annotation
General-purpose toolkits (<i>Biopython</i> , <i>pydna</i> , <i>SeqPro</i>) [7–9]	Manipulate sequences, simulate cloning, prepare arrays for modeling. Python packages (PyPI).	Parsing, transformation, and file-format support	Transformed sequences
Sequence optimization tools (<i>CodonGenie</i> , <i>DNA Chisel</i> , <i>ledidi</i>) [10–12]	Optimize individual sequences for codon usage, synthesis constraints, or a model’s objective. Python packages (PyPI); web services.	Degenerate codons covering a target residue set; hard constraints and scored objectives combined in one problem; gradient-based editing guided by a predictive model	An optimized or edited sequence, or a degenerate codon

Table 2 Capabilities of tools for designing DNA sequence libraries. ● denotes a capability the tool provides; ◐ one provided with restrictions; ○ one the tool does not provide. Capabilities are counted only where the tool provides a dedicated operation, parameter, or mode.

	DNA Chisel	MPRA Design Tools	MPRAnator	Mutation Maker	Oligopool Calculator	PoolParty	tangermeme	ValiAnT
<i>Library Specification</i>								
Library object	○	◐	○	◐	◐	●	◐	○
Composable operations	◐	○	◐	○	●	●	◐	○
Mixed variant types in one library	○	●	●	◐	○	●	○	●
<i>Variant Generation</i>								
Saturation mutagenesis	○	○	○	○	◐	●	●	●
Randomly sampled variants	●	◐	●	○	○	●	○	○
Pairwise and higher-order variants	●	○	●	●	◐	●	◐	○
Model-guided variants	●	○	○	○	○	◐	●	○
<i>Variant Types</i>								
Codon / amino-acid substitutions	●	○	○	◐	○	●	○	●
Insertions and deletions	○	●	◐	○	○	●	●	●
Combinatorial multi-motif placement	○	○	●	○	○	●	●	○
Recombination	○	○	○	○	○	●	○	○
Shuffling	○	○	◐	○	○	●	●	○
<i>Physical Construction</i>								
Synthesis-constraint checking	●	◐	◐	○	●	◐	○	◐
Constraint-based optimization	●	◐	◐	●	●	○	○	○
Primer design	◐	○	○	●	●	○	○	○
<i>Metadata and Inspection</i>								
Per-sequence design cards	●	●	◐	●	●	●	○	●
Automatic naming	○	◐	●	●	◐	●	◐	●
Sequence styling	○	○	◐	○	○	●	○	○
<i>Genomic Integration</i>								
Genome coordinates	○	◐	●	○	○	◐	◐	●
Transcript / annotation aware	○	○	○	○	○	○	○	●

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